A Performance-Centric Evaluation of Machine Learning (ML) and Deep Learning (DL) Models for Sentiment and Opinion Analysis

Prabal Sharma
Shoolini University
Solan, India
prabalsharma00@gmail.com

Anurag Rana
Shoolini University
Solan, India
anuragrana.anu@gmail.com

***Abstract*— Opinion mining, commonly known as sentiment analysis (SA), is crucial for the retrieval of useful knowledge from massive quantities of text that is written over various subjects as vast as web social media and customer opinions of products. In this paper, the authors assess performance for specific ML and DL architectures specifically adapted for sentiment and opinion mining tasks. This paper does an in-depth comparison of the standard models used in machine learning, such as SVM, Naive Bayes, and Decision Trees, with the more advanced deep-learning methodologies like LSTM networks, CNNs, and integrated architectures that merge both paradigms. The main goal is to evaluate these models' effectiveness in classifying sentiment into positive, negative, and neutral classes among diverse datasets that range from evaluating products to social media communications. This paper compares and evaluates the performance of the models based on key performance metrics such as accuracy, precision, recall, and F1-score. Based on a comparative analysis, this study identifies the strengths and weaknesses and the trade-offs that the ML and DL techniques carry, especially while dealing with noisy, complex datasets coming from real-world scenarios on a large scale. This brings to light the appropriateness of models for certain sentiment and opinion analysis tasks besides giving a clearer view regarding their applicability and limitations. Findings such as these point the way forward in enhancing the robustness, scalability, and generalization across models for myriad formats of text. Future Research Directions. This could push the frontiers of the research by rectifying the present inadequacies in terms of insensitivity to noise, adaptability of models, and efficiency of algorithms for the models.**

Keywords*— *Opinion mining, sentiment analysis, machine learning, deep learning, performance evaluation, natural language processing (NLP)

I. Introduction

Opinion Mining is the extraction of subjective information from text. Opinion mining, sometimes known as sentiment analysis (SA), has been defined to be the ability of recognizing and categorizing opinions held in textual data. The approach can identify that the sentiment is either negative or positive or neutral. This plays an important role in extracting meaningful insights from a lot of unstructured data generated each day across sites that include online communities, product review sites, and customer feedback sites. [1], [2]. Organizations and researchers use opinion mining to analyze the sentiment of people, improve the product, or inform policy [3]. Many computational models have been designed in the past for opinion mining. Traditional approaches adopted by (ML) towards such tasks as SA, or to be more precise, sentiment classification, include models like (SVM), Naive Bayes, and Decision Trees. [4]. Therefore, what is said on social media can be analysed to tell companies how they can improve consumer satisfaction, while public opinion monitoring on policies or any events gives decision-making insights.[5]. (ML) and (DL) have been the most popular opinion mining methods in recent years [6].

The rapid growth of online content has increased the demand for effective opinion mining, in which the precision of categorizing sentiment plays a pivotal role in collecting useful information and extracting actionable insights. This paper focuses on performance assessment of different ML and DL approaches in opinion mining [7].

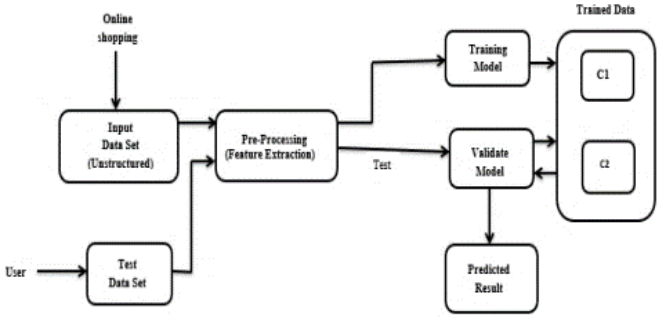

Fig. 1. Architecture for Opinion and (SA)

Here, different algorithms ranging from traditional ML classifiers to state-of-the-art neural networks will be studied to have a comparison in terms of their performance metrics regarding accuracy over different datasets. This study further extends the analysis with regard to the effect of different feature extraction and pre-processing techniques on model performance.

The highest-accuracy models are described within this paper that attained the most precision in analysing the sentiment and opinion, providing guidance to future insights that can be carried out with these models to solve real-world problems.

Authorized licensed use limited to: Indian Inst of Inform Technology Guwahati. Downloaded on April 10,2026 at 10:50:20 UTC from IEEE Xplore. Restrictions apply.

II. Literature Review

The Study "Opinion Mining and (SA)" discover the rapidly growing domain of opinion mining and (SA) computerized analysis of views and subjectivity in textual data. The review and blogs have been steadily on the rise online. More users demand systems that can evaluate and summarize subjective information accurately SA aims to grasp and group sentiments shown in text from social media. It uses many ML methods and lexicon-based techniques to spot feelings as negative, positive, or neutral [8]. In online shopping, analysing the mood of product reviews has become essential for businesses to grasp feedback and improve what they offer to buyers. Analyzing research work that has been done from 2018 to 2024 reflects that the usage of machine learning and deep learning is common. Thereby, the use of SVM and LSTM models gained high accuracy for tasks involving classification of sentiments.[9] Combining artificial intelligence with ML permits analysing massive volumes of unstructured data gathered through social networking sites, a customer feedback system, and polls for measuring and interpreting the sentiment of customers so that it helps in offering strategic insights for their refinement in offering [10].

ML and DL have proven powerful tools in healthcare for analysing huge amount of unstructured data. ML approaches help uncover useful information about how well drugs work and patient satisfaction via recognizing patterns and trends, healthcare providers may come to better choices, eventually enhancing treatment for patients and results. (ML) and (DL) become essential for understanding the patient experience and improving medical treatment techniques in data-driven world [6] [11]. Various traditional ML and DL models have been examined for their efficiency and effectiveness' in understanding drug efficacy using patient reviews. The models are SVM, XGBoost and Random Forest. Random Forest achieved accuracy at 97%, followed by SVM at 95% as well as by XGBoost at 94%. The outcome reflects that now traditional ML models have become more successful for (SA) in the context of the success of drugs. The importance of (ML) lies in the fact that it delivers understandable results and processes structured data efficiently, and hence it has become an essential tool in the field of healthcare data analysis [10]. This paper discusses various models of (ML) that are being applied in opinion mining, including Naive Bayes, SVM, and Logistic Regression., which has described feature extraction techniques such as TF-IDF and n-grams contributing to the proper classification of sentiments. The study further emphasizes the handling of noisy and unstructured data especially that derived from user-generated content like customer reviews. Results showed that SVM are effective but several algorithms integrated together in hybrid approach can better yield accuracy [11]. The study explores supervised ML techniques and focuses more on its applications in sentiment classification in text data sets. It explains the robustness of Random Forests, Decision Trees, and SVM models in dealing with vast data. The paper also explains the need for feature selection and preprocessing steps like stemming, or bringing words to their base form, and removal of redundant words. According to research results, a combination of (ML) models with proper preprocessing results in better accuracy of the sentiment categorization [4]. The study looks into analyzing customer online reviews by applying (ML) models, like Gradient Boosting, Random Forests, and SVM. Studies in this direction emphasize the key role of appropriate data cleaning and feature engineering to deal with noisy and duplicated data. The study points out that (ML) approaches provide organizations with the capability of automating the task of sentiment categorization for large-scale customer feedback [12]. SVM for (SA) of reviews of goods on e-commerce websites shown the efficiently categorize sentiments into positive and negative categories and preprocessing steps like tokenization have been crucial in improving model performance [9]. Integration of Fuzzy Logic with SVM for the SA of cosmetics product reviews that are dominated by subjective and varying customer feedback. The results showed that SVM achieved 90% accuracy whereas, fuzzy logic achieved slightly less at 88%. The strength of combining fuzzy logic with (ML) techniques is reported as very strong for product review analysis where the presence of subjective and varied customer feedback was there [13]. Their study focused on the importance of preprocessing techniques like choosing features and improving the accuracy of (ML) models for (SA) [14] analysis methods using Naive Bayes and SVM for categorizing online shopping reviews of goods. Their study compared the performance of both models and identified that SVM has a higher classification accuracy than Naive Bayes. The researchers highlighted the importance of preprocessing. With the help of Removing common words and shortening words to their basic form to enhance the precision of (ML) models for (SA) tasks [15]. SVM has been applied to sentiment extraction and compared to the advanced model, Bidirectional Long Short-Term Memory (BiLSTM).The combination of the BiLSTM-CNN resulted in an accuracy of 93%, outperforming SVM, which achieved an accuracy of 91%. The study presented a complementary role of the traditional ML model alongside more complex DL techniques [16]. In the product review, the ensemble methods combining the ML models like SVM and Random Forest proved to be useful for SA. These methods used strengths of the individual classifiers, predicting better overall accuracy [17].

Advanced (DL) algorithms, including CNN-LSTM, BiLSTM and XLNet have demonstrated superior capabilities in opinion mining, particularly for movie review. These models combine the ability of CNN to capture local dependencies with LSTM's strength in modeling global sequential patterns. Studies on the IMDb dataset achieve great accuracy and highlight the better results achieved by (DL) models in handling complex and unstructured reviews compared to standard techniques [18]. Using CNN for (SA) for movie reviews and achieving high accuracy by getting relevant features from text. A hybrid model merging BiLSTM and CNN for (SA) and the model reaching an accuracy of 93% with the hybrid model as well as 91% with BiLSTM [19]. Sentiment analysis for Chinese e-commerce product evaluations is conducted using a transformer-based model that combines RCNN and Text CNN, as implemented in ERNIE. The model attained a fantastic accuracy of 95.4% and showing the power of transformer-based (DL) models for several languages' (SA) [20]. The combination of CNN for feature extraction and LSTM for learning sequences has proved very effective in dealing with vast datasets.

Authorized licensed use limited to: Indian Inst of Inform Technology Guwahati. Downloaded on April 10,2026 at 10:50:20 UTC from IEEE Xplore. Restrictions apply.

TABLE I. SUMMARY OF REVIEWED LITERATURE IN TERMS OF ML AND DL TECHNIQUES' PERFORMANCES

Ref.	Dataset	Preprocessing	Techniques	Model	Performance
[18]	IMDb Movie Reviews	Stopword removal, lemmatization, tokenization	CNN, LSTM, XLNet	CNN-LSTM,XLNet	Accuracy: 93.74%
[25]	Customer Reviews	Tokenization, stemming, stopword removal	Naive Bayes, SVM	SVM	Moderate Accuracy
[1]	Twitter Data	Emoji lexicon mapping, tokenization	CNN, LSTM, Emoji Sentiment Lexicon	CNN-LSTM	Accuracy: 83.7%
[2]	Twitter Data	Tokenization, hashtag cleanup	LSTM, CNN	LSTM, CNN	High Accuracy
[3]	Customer Reviews	Tokenization, stemming	Clustering, SVM	SVM, K-Means	Good Performance
[28]	Customer Feedback	Tokenization, stopword removal	SVM, Naive Bayes, Decision Trees	SVM	Good Accuracy
[30]	Course Reviews	Tokenization, lemmatization	LSTM, Word Embeddings	LSTM	High Accuracy
[31]	Online Customer Reviews	Tokenization, sentiment lexicon integration	Naive Bayes, SVM, Gradient Boosting	Gradient Boosting	High Accuracy
[32]	Dataset of 2,229 product reviews	Tokenization, stopword removal	SVM	SVM	High accuracy
[33]	10,000 skincare product reviews	Sentiment lexicon-based analysis	Fuzzy Logic, SVM	SVM, Fuzzy Logic	SVM: 90%, Fuzzy Logic: 88%
[34]	15,000 e-commerce reviews	Lemmatization, stopword removal	Naive Bayes, SVM	SVM, Naïve Bayes	SVM outperformed Naive Bayes
[35]	5,000 e-commerce reviews	Tokenization, feature selection	SVM	SVM	High accuracy
[36]	30,000 product reviews	Stemming, punctuation removal	SVM, Random Forest	SVM, Random Forest	High accuracy
[37]	30,000 Amazon reviews	Text normalization, feature selection	CNN-LSTM	CNN-LSTM	92% accuracy
[38]	40,000 online reviews	Lemmatization, stopword removal	LSTM, GRU	LSTM, GRU	LSTM: 87%, GRU: 85%
[39]	60,000 product reviews	Attention mechanism	BiLSTM with Attention	BiLSTM-Attention	93% accuracy

The results of analyzing more than 30,000 Amazon product reviews using a hybrid CNN-LSTM model achieved the highest accuracy of 92%. This hybrid approach learns local patterns and sequential dependencies, rendering it a strong candidate for large-scale SA [21],[22]. Studies on Gated Recurrent Units (GRUs) and Bidirectional GRUs (BiGRUs) were originally applied to the (SA) of the agricultural products. Both models were solid enough to deal with the sequential data and most accurate of BiGRUs was 91%. Most importantly, this research pointed out the attentiveness and effectiveness of using the attention mechanisms in lifting the performance of the model [23]. A BiLSTM hybrid model was also used in the multilingual dataset, proving to be robust for SA tasks in various languages. Pre-processing steps, tokenization, and embedding using already-trained word vectors contribute to the adaptability of the model to various datasets and applications [24]. Table 1 summarizes the reviewed literature.

III. DISCUSSION

The performance analysis of (ML) and (DL) models in opinion mining for opinion and (SA) is done in detail with the strengths and weaknesses of the respective models. Among ML models, Naive Bayes, SVM, and Random Forest have been excellent while analyzing structured data. They achieve high accuracy using certain pre-processing techniques, such as feature extraction and tokenization, for example, TF-IDF, n-grams. For instance, if the application procedure were done correctly, the accuracy level of SVM could be up to 91%. But these models aren't optimal for the task of processing complex, unstructured data like that from social media or consumer reviews. However, models of DL like CNN and LSTM can very well handle massive data input as they expose relationships that a human's brain is hard to catch a glimpse of. Hybrid models, such as CNN-LSTM, perform better than their isolated counterparts. Accuracy rates of up to 92% to 95% are achieved. High computing cost and overfitting are still present in DL models. Future studies should focus on scalability, generalization of models, and use of multimodal data for more accurate sentiment prediction across various applications. The Table 2 shows the metrics performance of (ML) models Naive Bayes, Random Forest, SVM and the DL model (CNN-LSTM). ML Models Naive Bayes, Random Forest and SVM perform well with structured data and are faster to train but may struggle with noisy and unstructured data like social media and movie reviews. DL Models (CNN-LSTM) achieve better performance by capturing complex patterns in unstructured data but come with the trade-off of higher computational cost and training time.

TABLE II. PERFORMANCE METRICS OF THE ML AND THE DL MODELS

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.8391	0.8463	0.8287	0.8374
CNN-LSTM	0.8729	0.8863	0.8556	0.8707
Random Forest	0.8340	0.8438	0.8200	0.8315
SVM	0.8721	0.8694	0.8758	0.8726

To perform a compared study of (ML) and (DL) models for (SA) (as discussed in the research), we will use the Scikit-learn library for traditional (ML) models such as (Naive Bayes,SVM and Random Forest) and Keras/TensorFlow for DL models (such as CNN-LSTM). We will assess the models using performance indicators like accuracy, recall, precision, and F1-score. Table 3, movie reviews dataset, is an example containing reviews with sentiment labels; Positive or Negative. The labeled examples are used in training models that predict sentiments in new reviews.

Authorized licensed use limited to: Indian Inst of Inform Technology Guwahati. Downloaded on April 10,2026 at 10:50:20 UTC from IEEE Xplore. Restrictions apply.

TABLE III. MOVIE REVIEWS DATASET

Review	Sentiment
Amazing movie, great plot and acting! Highly recommend!	Positive
Boring story, poor acting, and bad special effects. Not worth watching.	Negative
Heartwarming story with superb acting. Very emotional.	Positive
Complete disaster, predictable plot, and bad acting.	Negative
Incredible soundtrack and performances. Story kept me hooked.	Positive
Terrible movie, forced jokes, and bad pacing. Don't watch it.	Negative
Surprised by how good this movie was. Excellent cinematography.	Positive
Dull and uninteresting. The plot dragged on too long.	Negative
A must-watch! The direction and acting were top-notch. A true masterpiece.	Positive

The process involves loading and preprocessing of the data, Initially, the IMDB dataset is received and tokenized, before the sequences are padded to make consistent inputs. Vectorizing the Data using TF-IDF, a technique very commonly found in numerical data, text data transformed into a numerical representation that was utilized for linear models most often. Training of models including traditional ML algorithms and a DL model, are trained on the dataset. After the training process, different models are judged by several parameters: precision, recall, precision, accuracy, and F1-score. The outcome metrics of all models are shown for comparison.

IV. RESULTS AND DISCUSSIONS

A. Mathematical Equations

General Sentiment Classification Formulation for Multiclass (e.g., Positive, Negative, Neutral) and Binary (e.g., Positive vs. Negative) Classification explain in equations 1 and 2.

$$\hat{y} = sigmoid(w.x + b) \quad (1)$$

$$\hat{y} = softmax(w.x + b) \quad (2)$$

B. Cross-Entropy Loss for Multiclass

Classification: Cross-Entropy Loss is utilized to assess the performance of a classification model for multiclass classification explain in equation 3. In this equation C is the number of classes, $y_{i,j}$ is the true label (one-hot encoded) and $\hat{y}_{i,j}$ is the predicted probability for class j. This function of loss helps reduce the difference between the expected probabilities and the actual tags.

$$\mathcal{L}(x_i, y_i) = -\sum_{j=0}^{C-1} y_{i,j} \log(\hat{y}_{i,j}) \quad (3)$$

ML and DL model for sentiment classifications: SVM seeks to locate the hyperplane that separates various classes with the biggest margin in equation 4.

$$\min_w \frac{1}{2}\|w\|^2 = subject\ to\ y_i(w.x_i + b) \geq 1 \quad (4)$$

Naive Bayes calculates the likely outcome of each class provided the input features using Bayes' theorem as expressed in equation 5.

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} \quad (5)$$

The CNN convolution operation on the input sequence to extract local features using equation 6.

$$y = Conv(x, W) + b \quad (6)$$

The LSTM captures long-term dependencies in sequences using gates (forget, input, and output gates) clearly explained in equation 7 and 8. The CNN-LSTM, Combines CNNs for local features and LSTMs for sequential dependencies to improve sentiment prediction.

$$f_t = \sigma(W_f . [h_{t-1}, x_t] + b_f) \quad (7)$$

$$h_t = o_t * \tanh(C_t) \quad (8)$$

C. Comparative Study Analysis

The study compares how good Naive Bayes, SVM, Random Forest, and CNN-LSTM performed on important measures such as Accuracy, Precision, Recall, and F1-Score. It also looks at how the model pairs performed together by comparing their average scores for each measure to see which one is better. SVM always has good and balanced performance among all the metrics, doing well on Recall, and thus would be a best choice in applications where sensitivity needs to be high. CNN-LSTM has high Precision and Accuracy, performing well at making very accurate predictions with low false positives; it is most fitting for tasks involving (SA).

Naive Bayes has the worst overall performance, although computationally efficient; it can be used only for lightweight tasks where speed is very important. Random Forest exhibits moderate performance, but it falls short compared to SVM and CNN-LSTM most of the time.

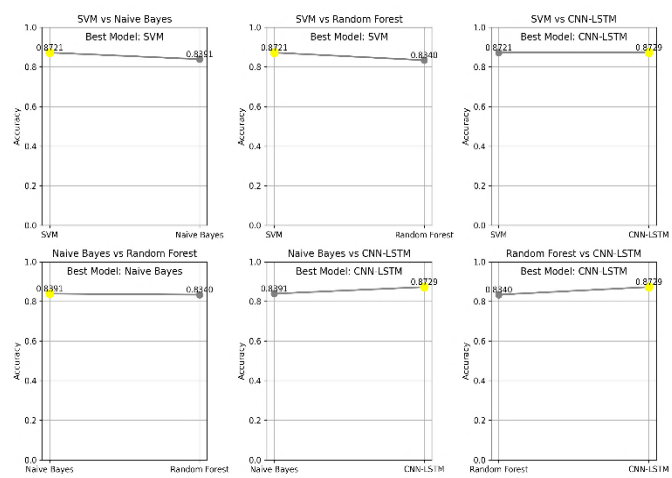

Fig. 2. Accuracy evaluation of model pairs

In Figure 2, model pairs are compared according to the Accuracy metric. Each of the pairs contains two models, and the goal is to identify which model performs better in terms of accuracy. The top-performing model in each comparison is highlighted.

Authorized licensed use limited to: Indian Inst of Inform Technology Guwahati. Downloaded on April 10,2026 at 10:50:20 UTC from IEEE Xplore. Restrictions apply.

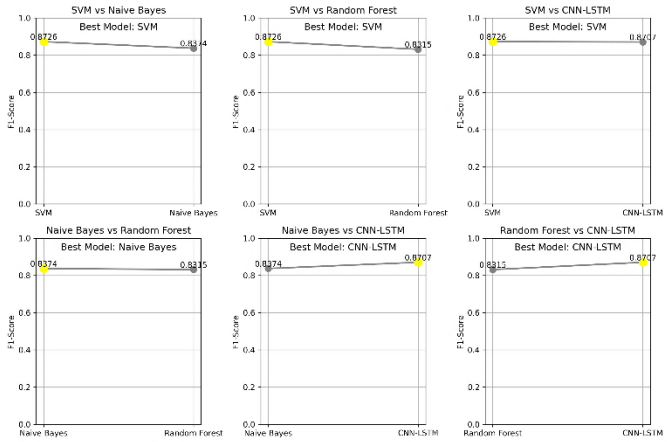

Fig. 3. The Model Pair Precision Evaluation

In figure 3, model pairs are compared according to the precision metric. Each of the pairs contains two models, and the goal is to identify which model performs better in terms of precision. The top-performing model in each comparison is highlighted. In figure 4, model pairs are compared according to the Recall metric. Each of the pairs contains two models, and the goal is to identify which model performs better in terms of Recall. The top-performing model in each comparison is highlighted.

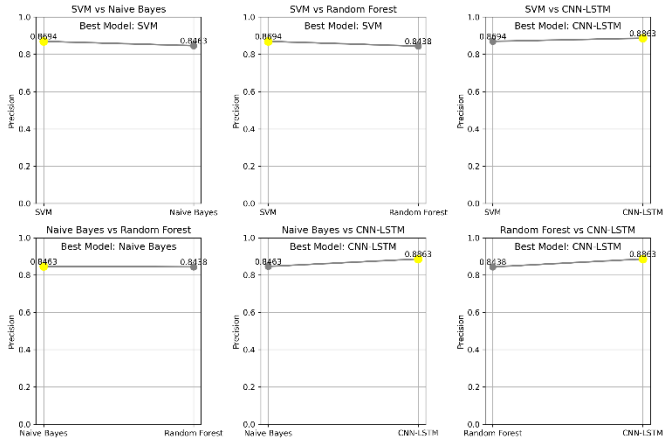

Fig. 4. The Recall of Model Pairs

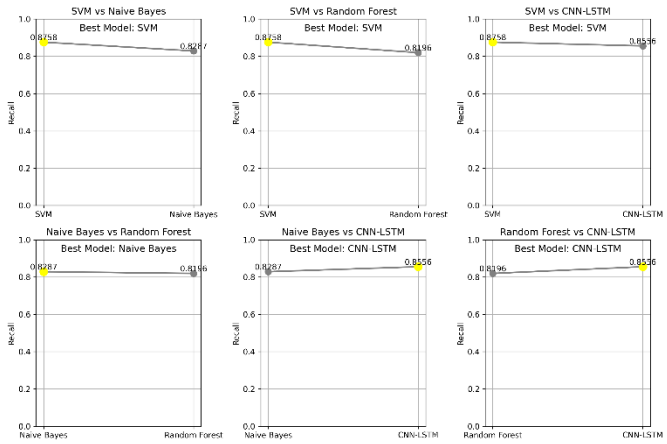

Fig. 5. The model pair Precision evaluation

In Figure 5, model pairs are compared according to the F1-Score metric. Each of the pairs contains two models, and the goal is to identify which model performs better in terms of F1-Score. The top-performing model in each comparison is highlighted.

When comparing pairs based on average scores of Naive Bayes, SVM and Random Forest, the SVM outperforms over Random Forest and Native Bayes. But CNN-LSTM is significantly performing better than SVM, Naive Bayes and Random Forest. Figure 6 depicts the average scores of model pairs' performance across the Precision, Accuracy, Recall, and F1-Score metrics using bar plots for direct visual comparison.

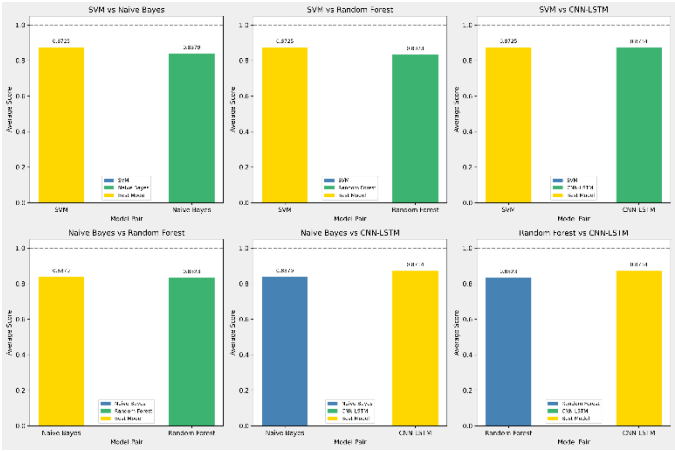

Fig. 6. The Average Score of Traditional ML\DL and LLM

V. Conclusion

In conclusion, CNN-LSTM outperformed all the models in average performance and was superior in Accuracy and Precision. Therefore, CNN-LSTM is best for applications that require high accuracy. SVM is best in Recall and hence best suited for applications where recall is critical. The Random Forest offered balanced performance, whereas Naive Bayes is the best for lightweight applications but generally performs poorly. In most of the applications, CNN-LSTM is preferred, and SVM is preferred in recall-focused applications.

Future study should focus on improving the scale, speed, and generalizability of these models. The integration of multimodal data (e.g., text, images, and emojis) into (SA) frameworks offers significant potential for improved accuracy and robustness. Additionally, adopting transfer learning techniques and developing lightweight models could mitigate computational challenges, enabling faster and more practical applications across diverse fields. These advancements would pave the way for more accessible, efficient, and impactful machine-driven (SA) solutions.

References

[1] C. Messaoudi, Z. Guessoum, and L. ben Romdhane, “A (DL) Model for Opinion mining in Twitter Combining Text and Emojis,” in Procedia Computer Science, Elsevier B.V., 2022, pp. 2628–2637. doi: 10.1016/j.procs. 2022.09.321.

[2] A. Alsaeedi and M. Z. Khan, “A Study on (SA) Techniques of Twitter Data,” 2019. [Online]. Available: www.ijacsa.thesai.org.

[3] S. Moza, E. Ashraf, H. Agrawal, and Vikash, “Customer Opinion Analysis and Topic Extraction Using (ML),” in ACM International

Authorized licensed use limited to: Indian Inst of Inform Technology Guwahati. Downloaded on April 10,2026 at 10:50:20 UTC from IEEE Xplore. Restrictions apply.

Conference Proceeding Series, Association for Computing Machinery, Oct. 2024, pp. 502–507. doi: 10.1145/3675888.3676104.

[4] P. Kumar Lakineni AssocProf and Nv. Reddy Assistant, "Opinion Mining using ML Approaches," Mukt Shabd Journal, vol. 11(4), 2020.

[5] L. Zhang, S. Wang, and B. Liu, "Deep learning for sentiment analysis: A survey", 2018, doi: 10.1002/widm.1253.

[6] Nikhil Sanjay Suryawanshi, "SA with ML and DL: A survey of techniques and applications," International Journal of Science and Research Archive, vol. 12, no. 2, pp. 005–015, Jul. 2024, doi: 10.30574/ijsra.2024.12.2.1205.

[7] I. R. Vanani and M. Amirhosseini, "DL for Opinion Mining," 2018, pp. 40–65. doi: 10.4018/978-1-5225-6117-0.ch003.

[8] A. A. Q. Aqlan, B. Manjula, and R. Lakshman Naik, "A study of SA: Concepts, techniques, and challenges," in Lecture Notes on Data Engineering and Communications Technologies, vol. 28, Springer Science and Business Media Deutschland GmbH, 2019, pp. 147–162. doi: 10.1007/978-981-13-6459-4_16.

[9] A. Daza, N. D. González Rueda, M. S. Aguilar Sánchez, W. F. Robles Espíritu, and M. E. Chauca Quiñones, "SA on E-Commerce Product Reviews Using ML and DL Algorithms: A Bibliometric Analysisand Systematic Literature Review, Challenges and Future Works," Nov. 01, 2024, Elsevier B.V. doi: 10.1016/j.jjimei.2024.100267.

[10] B. O. Nwogu, "Comparative Assessment of ML And Comparative Assessment of ML and DL Models For Drug Effectiveness Using, 2024. [Online]. Available: https://scholarworks.lib .csusb.edu/etd

[11] V. K. Solanki, N. H. H. Cuong, and Zonghyu (Joan) Lu, "Opinion Mining," 2018, pp. 66–82. doi: 10.4018/978-1-5225-6117-0.ch004.

[12] B. Kanwal et al., "Opinion Mining from Online Travel Reviews: An Exploratory Investigation on Pakistan Major Online Travel Services Using Natural Language Processing," IEEE Access, vol. 11, pp. 29934–29945, 2023, doi: 10.1109/ACCESS.2023.3260114.

[13] B.M, Meghana K S, Shivani S Raykar, Spandana M K, and Vinutha V, "Sentimental Analysis on Cosmetics using ML," ACS Journal for Science and Engineering, vol. 2, no. 2, 2022, doi: 10.34293/acsjse. v2i2.31.

[14] I. M. Yulietha, S. A. Faraby, Adiwijaya, and W. C. Widyaningtyas, "An implementation of support vector machine on sentiment classification of movie reviews," in Journal of Physics: Conference Series, 2018. doi: 10.1088/1742-6596/971/1/012056.

[15] A. M. Rahat, A. Kahir, and A. K. M. Masum, "Comparison of Naive Bayes and SVM Algorithm based on (SA) Using Review Dataset," in Proceedings of the 2019 8th International Conference on System Modeling and Advancement in Research Trends, SMART 2019, 2020. doi: 10.1109/SMART46866.2019.9117512.

[16] H. Fang, G. Jiang, and D. Li, "(SA) based on Chinese BERT and fused deep neural networks for sentence-level Chinese e-commerce product reviews," Systems Science and Control Engineering, vol. 10, no. 1, 2022, doi: 10.1080/21642583.2022.2123060.

[17] T. Ahmed Khan, R. Sadiq, Z. Shahid, M. M. Alam, and M. Mohd Su'ud, "(SA) using Support Vector Machine and Random Forest," Journal of Informatics and Web Engineering, vol. 3, no. 1, 2024, doi: 10.33093/jiwe.2024.3.1.5.

[18] M. M. Danyal, M. Haseeb, S. S. Khan, B. Khan, and S. Ullah, "Opinion Mining on Movie Reviews Based on (DL) Models," Journal on Artificial Intelligence, vol. 6, no. 1, pp. 23–42, 2024, doi: 10.32604/jai.2023. 045617.

[19] M. Mishra and A. Patil, "Sentiment Prediction of IMDb Movie Reviews Using CNN-LSTM Approach," in 2023 International Conference on Control, Communication and Computing, ICCC 2023, 2023. doi: 10.1109/ICCC57789.2023.10165155.

[20] W. Huang, M. Lin, and Y. Wang, "(SA) of Chinese E-Commerce Product Reviews Using ERNIE Word Embedding and Attention Mechanism," Applied Sciences (Switzerland), vol. 12, no. 14, 2022, doi: 10.3390/app12147182.

[21] L. Wang, K. Tan, and Y. Huang, "Reputation Analysis of E-commerce Products Based on Online Reviews—Take Amazon as an Example," J Econ Public Financ, vol. 6, no. 2, 2020, doi: 10.22158/jepf.v6n2p128.

[22] S. Ahmad, S. M. Saqib, and A. H. Syed, "CNN and LSTM based hybrid DL model for SA on Arabic text reviews," Mehran University Research Journal of Engineering and Technology, vol. 43, no. 2, p. 183, Apr. 2024, doi: 10.22581/muet1982.3130.

[23] Z. Zhou, J. Chen, J. Wu, and R. Wang, "SA Method for Agricultural Product Review Based on Corpus Characteristics and (DL) Model," in Proceedings - 2022 International Conference on (ML) and Intelligent Systems Engineering, MLISE 2022, 2022. doi: 10.1109/MLISE57402. 2022.00091.

[24] M. Wankhade, C. S. R. Annavarapu, and A. Abraham, "CBMAFM: CNN-BiLSTM Multi-Attention Fusion Mechanism for sentiment classification," Multimed Tools Appl, vol. 83, no. 17, 2024, doi: 10.1007/s11042-023-17437-9.

[25] S. Rana and A. Singh, "Comparative analysis of sentiment orientation using SVM and Naive Bayes techniques," in Proceedings on 2016 2nd International Conference on Next Generation Computing Technologies, NGCT 2016, 2017. doi: 10.1109/NGCT.2016.7877399.

[26] K. Smagulova and A. P. James, "A survey on LSTM memristive neural network architectures and applications," 2019. doi: 10.1140/epjst/e2019-900046-x.

[27] Y. Agarwal, R. Katarya, and D. K. Sharma, "DL for Opinion Mining: A Systematic Survey," in 2019 4th International Conference on Information Systems and Computer Networks, ISCON 2019, 2019. doi: 10.1109/ISCON47742.2019.9036187.

[28] P. Kumar Lakineni AssocProf and Nv. Reddy Assistant, "Opinion Mining Using (ML) Approaches."

[29] S. Sukheja, S. Chopra, and M. Vijayalakshmi, "Sentiment Analysis using Deep Learning - A survey," 2020 International Conference on Computer Science, Engineering and Applications (ICCSEA), Gunupur, India, 2020, pp. 1-4.

[30] A. Koufakou, "(DL) for opinion mining and topic classification of course reviews," Educ Inf Technol (Dordr), vol. 29, no. 3, pp. 2973–2997, Feb. 2024, doi: 10.1007/s10639-023-11736-2.

[31] T. K. Phung, N. A. Te, and T. T. T. Ha, "A (ML) approach for opinion mining online customer reviews," in Proceedings - 2021 21st ACIS International Semi-Virtual Winter Conference on Software Engineering, Artificial Intelligence, Jan. 2021, pp. 243–246. doi: 10.1109/SNPD Winter52325.2021. 00059.

[32] Y. Basani, H. V. Sibuea, S. Ida Patona Sianipar, and J. Presly Samosir, "Application of (SA) on Product Review E-Commerce," in Journal of Physics: Conference Series, 2019. doi: 10.1088/1742-6596/1175/1/ 012103.

[33] R. Liang and J. qiang Wang, "A Linguistic Intuitionistic Cloud Decision Support Model with (SA) for Product Selection in E-commerce," International Journal of Fuzzy Systems, vol. 21, no. 3, 2019, doi: 10.1007/s40815-019-00606-0.

[34] R. V. Karthik and S. Ganapathy, "A fuzzy recommendation system for predicting the customers interests using (SA) and ontology in e-commerce," Appl Soft Comput, vol. 108, 2021, doi: 10.1016/j.asoc. 2021.107396.

[35] S. Ma, Y. Chai, and H. Zhang, "Rise of Cross-border E-commerce Exports in China," China and World Economy, vol. 26, no. 3, 2018, doi: 10.1111/cwe.12243.

[36] R. Punithavathi, M. Premkumar, G. Sinha, and K. Chandrasekaran, "Transforming (SA) for e-commerce product reviews: Hybrid (DL) model with an innovative term weighting and feature selection," Inf Process Manag, vol. 61, no. 3, 2024, doi: 10.1016/j.ipm.2024.103654.

[37] H. J. Alantari, I. S. Currim, Y. Deng, and S. Singh, "An empirical comparison of (ML) methods for text-based (SA) of online consumer reviews," International Journal of Research in Marketing, vol. 39, no. 1, 2022, doi: 10.1016/j.ijresmar.2021.10.011.

[38] S. Sanyala and M. W. Hisamb, "Factors Affecting Customer Satisfaction with Ecommerce Websites - An Omani Perspective," in Proceeding of 2019 International Conference on Digitization: Landscaping Artificial Intelligence, ICD 2019, 2019. doi: 10.1109/ICD47981.2019.9105780.

[39] J. Zhang, A. Zhang, D. Liu, and Y. Bian, "Customer preferences extraction for air purifiers based on fine-grained (SA) of online reviews," Knowl Based Syst, vol. 228, 2021, doi: 10.1016/ j.knosys.2021.107259.

Authorized licensed use limited to: Indian Inst of Inform Technology Guwahati. Downloaded on April 10,2026 at 10:50:20 UTC from IEEE Xplore. Restrictions apply.